

EE 115 Lab 5

Frequency Modulation Truncation and Bandwidth

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Given FM signal

The passband FM signal is

$$u_{\text{FM}}(t) = \cos\left(2\pi f_c t + 2\pi k_f \int_0^t m(\tau) d\tau\right) \quad (1)$$

with modulation index $\beta = \frac{\Delta f_{\text{max}}}{B_m} = \frac{k_f \max |m(t)|}{B_m}$. For the single-tone message $m(t) = \cos(2\pi f_m t)$ (so $B_m = f_m$), the complex envelope is

$$u(t) = e^{j\beta \sin(2\pi f_m t)} = \sum_{n=-\infty}^{\infty} J_n(\beta) e^{j2\pi f_m n t} \quad (2)$$

where $J_n(\cdot)$ is the Bessel function of the first kind. Truncating the series to $|n| \leq K$ gives

$$u_K(t) = \sum_{n=-K}^K J_n(\beta) e^{j2\pi f_m n t}, \quad (3)$$

whose power over one message period is

$$P_{u_K} = J_0^2(\beta) + 2 \sum_{n=1}^K J_n^2(\beta). \quad (4)$$

Because $u(t)$ has unit power, $1 - P_{u_K}$ is the truncation power error.

1. Numerical evaluation and plots

Method: For each $\beta \in \{0.5, 2, 5, 10\}$ and $K = 0, \dots, 50$, compute P_{u_K} using (4), then $1 - P_{u_K}$. Figures 1–2 show the results on linear and dB scales (dB plotted as $10 \log_{10}(1 - P_{u_K})$ with a small floor to avoid $\log(0)$).

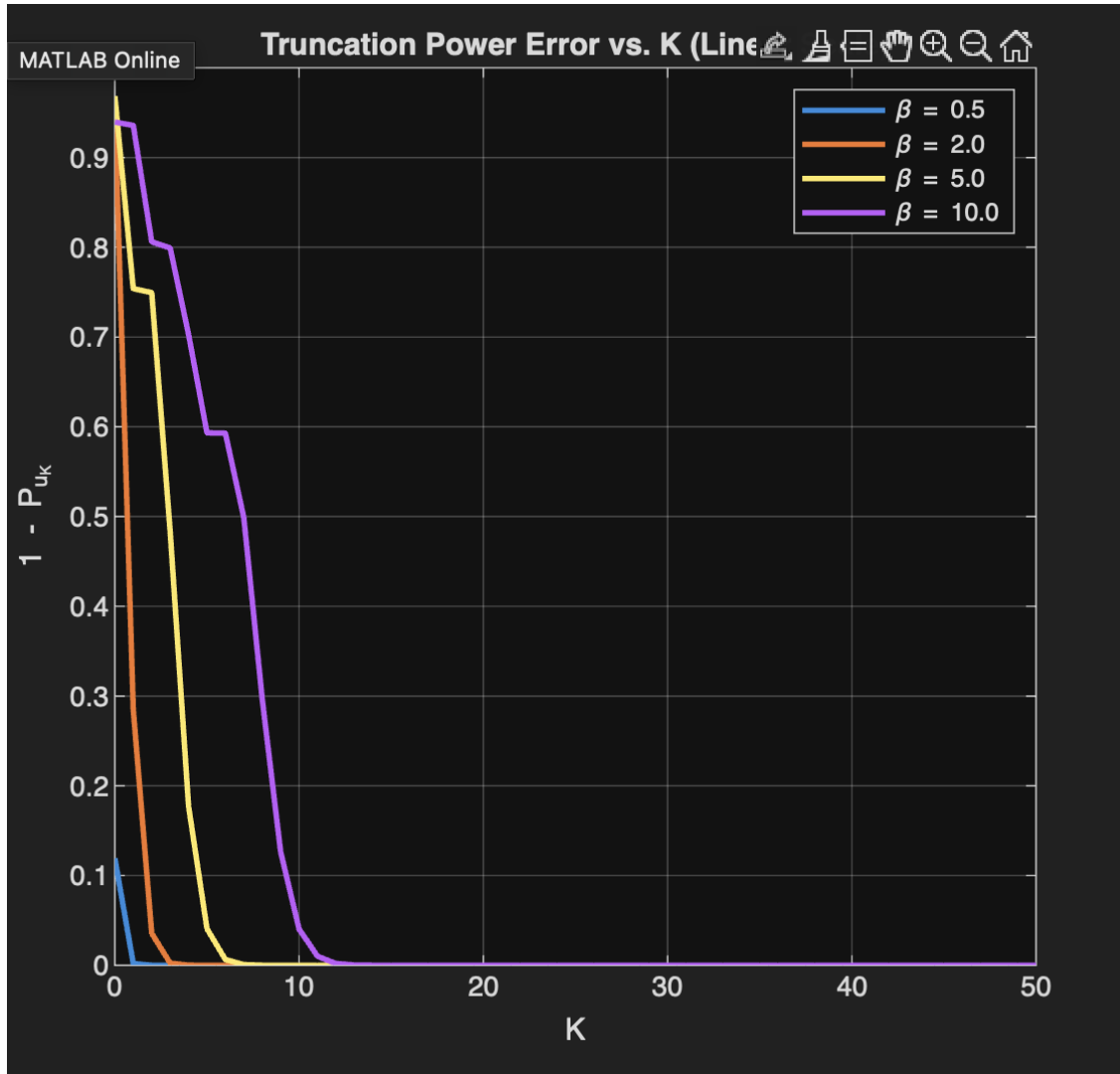


Figure 1: $1 - P_{u_K}$ vs. K (linear scale). Higher β requires larger K for the same residual error.

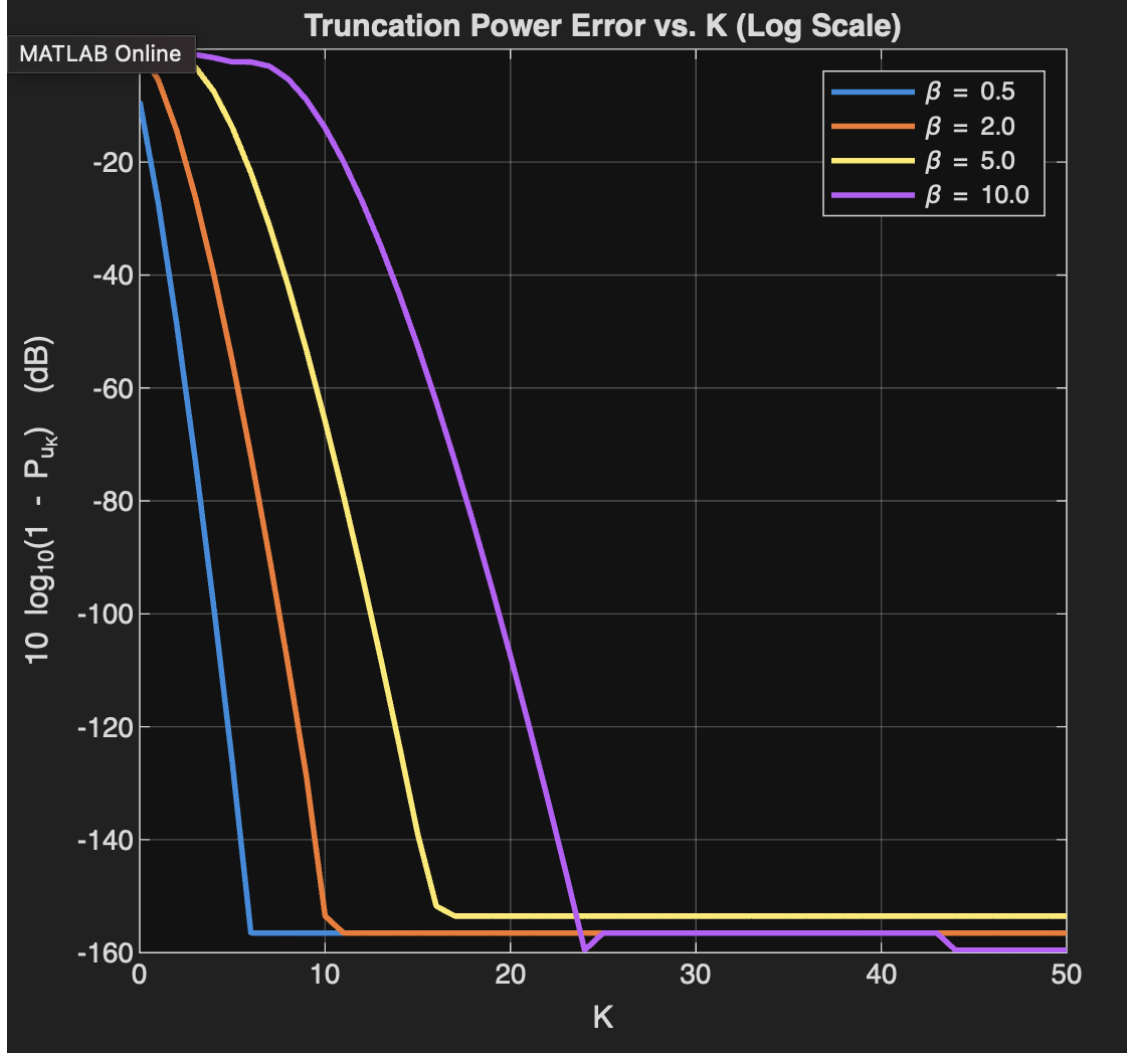


Figure 2: $10 \log_{10}(1 - P_{u_K})$ vs. K (dB). Residual power falls rapidly once $K \gtrsim \beta + 1$, consistent with Carson's rule.

How large must K be? Using the same computation, the first K that makes $1 - P_{u_K}$ drop below several thresholds is:

β	$1 - P_{u_K} \leq 10^{-2}$	$\leq 10^{-3}$	$\leq 10^{-6}$	$\leq 10^{-12}$
0.5	1	2	3	5
2.0	3	4	6	9
5.0	6	7	10	14
10.0	12	13	16	22

The table matches the visual takeaway: meaningful coefficients cluster around $n \approx \beta$, and setting $K \approx \beta + 1$ captures essentially all the power.

2. Why $1 - P_{u_K}$ is the error power

Define the truncation error $e(t) = u(t) - u_K(t)$. From (2),

$$e(t) = \sum_{|n| > K} J_n(\beta) e^{j2\pi f_m n t}. \quad (5)$$

The complex exponentials $e^{j2\pi f_m n t}$ are orthogonal over any integer multiple of the message period $T_m = 1/f_m$, so all cross terms vanish when computing power:

$$P_e = f_m \int_{-1/(2f_m)}^{1/(2f_m)} |e(t)|^2 dt \quad (6)$$

$$= 2 \sum_{n=K+1}^{\infty} J_n^2(\beta) \quad (\text{equal power for } \pm n) \quad (7)$$

$$= 1 - [J_0^2(\beta) + 2 \sum_{n=1}^K J_n^2(\beta)] = 1 - P_{u_K}. \quad (8)$$

Because the total power of $u(t)$ is 1, the omitted coefficients account exactly for the residual power. Hence $1 - P_{u_K}$ is the power of the truncation error $e(t)$.

3. Why $B_u \approx f_m(\beta + 1)$

The spectrum of the complex envelope (2) contains discrete lines spaced by f_m at frequencies $\pm n f_m$ with weights $J_n(\beta)$. For FM, $J_n(\beta)$ is negligible for $|n| > \beta + 1$ (the main lobe of J_n sits near $n \approx \beta$ and decays quickly outside that window). Truncating at $|n| \leq K \approx \beta + 1$ therefore retains essentially all the power, so the one-sided lowpass bandwidth of $u(t)$ is

$$B_u \approx K f_m \approx f_m(\beta + 1).$$

This matches Carson's rule: the passband FM bandwidth is $W \approx 2(B_m + \Delta f_{\max}) = 2f_m(\beta + 1)$, so the analytic baseband (complex envelope) occupies half that, $B_u \approx f_m(\beta + 1)$.

MATLAB/Octave code (lab_5.m)

The following script reproduces the figures and table above.

```
% EE 115 Lab 5 - FM truncation error
clear; close all; clc;

betas = [0.5 2 5 10];
Kmax = 50;
K = 0:Kmax;
err = zeros(numel(betas), numel(K));

for idx = 1:numel(betas)
    beta = betas(idx);
    J = besselj(0:Kmax, beta); % J_n for n = 0..Kmax
    for k = 0:Kmax
        power = J(1)^2 + 2*sum(J(2:k+1).^2); % P_{u_K}
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        err(idx, k+1) = 1 - power;           % truncation power error
    end
end

% Linear plot
figure;
hold on; grid on;
colors = lines(numel(betas));
for idx = 1:numel(betas)
    plot(K, err(idx, :), 'LineWidth', 2, 'Color', colors(idx, :));
end
xlabel('K'); ylabel('1-P_{u_K}');
title('Truncation_Power_Error_vs_K_Linear');
legend('\beta=0.5', '\beta=2.0', '\beta=5.0', '\beta=10.0', 'Location',
    'northeast');
set(gca, 'YLim', [0 1]); % keeps the y-axis focused
exportgraphics(gcf, 'image1.png', 'Resolution', 300);

% dB plot
figure;
hold on; grid on;
floorVal = eps; % avoid log of zero
for idx = 1:numel(betas)
    plot(K, 10*log10(max(err(idx, :), floorVal)), ...
        'LineWidth', 2, 'Color', colors(idx, :));
end
xlabel('K'); ylabel('10*log_{10}(1-P_{u_K})(dB)');
title('Truncation_Power_Error_vs_K_LogScale');
legend('\beta=0.5', '\beta=2.0', '\beta=5.0', '\beta=10.0', 'Location',
    'northeast');
set(gca, 'YLim', [-160 0]);
exportgraphics(gcf, 'image2.png', 'Resolution', 300);

% Optional: print thresholds used in the table
thresholds = [1e-2 1e-3 1e-6 1e-12];
fprintf('beta_K(@<=1e-2)_K(@<=1e-3)_K(@<=1e-6)_K(@<=1e-12)\n');
for idx = 1:numel(betas)
    line = zeros(1, numel(thresholds));
    for t = 1:numel(thresholds)
        kHit = find(err(idx, :) <= thresholds(t), 1) - 1; % zero-based K
        line(t) = kHit;
    end
    fprintf('%4.1f_%11d_%11d_%11d_%12d\n', betas(idx), line);
end
end

```